Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Preclinical Development of a Cordyceps militaris–Derived Botanical Drug for Respiratory Diseases and Lung Cancer			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Natural Pharmaceuticals	Bio Process/Equipment	Bio Process Technology
Concept of Proposed Technology	The proposed technology focuses on the **global preclinical development of a botanical drug candidate derived from *Cordyceps militaris*** and/or a **cordycepin-based natural product–derived synthetic drug** targeting **respiratory diseases, including chronic obstructive pulmonary disease (COPD), asthma, and lung cancer**. *Cordyceps militaris* has been shown to exhibit **anti-inflammatory, immunomodulatory, and anticancer activities**, with cordycepin identified as a key bioactive component. The project aims to translate these properties into a **prescription-grade pharmaceutical product** through systematic preclinical development aligned with regulatory expectations. The technology encompasses: • Standardization of raw materials and manufacturing processes • Quantitative control of marker compounds, including cordycepin • Mechanistic elucidation of multi-component pharmacological effects • Generation of globally acceptable nonclinical efficacy and safety data			

Rationale for Joint R&D Project	**Regulatory and Scientific Barriers**
	
	Achieving FDA IND clearance for botanical drugs presents significant regulatory and scientific challenges, particularly for small and medium-sized enterprises. These challenges include compliance with the FDA's **Botanical Drug Development Guidance**, generation of decision-grade nonclinical data, and demonstration of consistent product quality for multi-component natural products.
	
	Necessity of International Collaboration
	
	Joint R&D with an overseas research institute is essential to address these challenges by providing:
	
	1. **Regulatory Science Expertise**
	  ○ Support for nonclinical study design aligned with FDA botanical and synthetic drug guidance
	  ○ Consultation on IND dossier structure, content, and submission strategy
	2. **Advanced Efficacy Evaluation Models**
	  ○ Access to clinically relevant animal models for respiratory diseases and lung cancer
	  ○ Generation of globally credible efficacy data suitable for regulatory review
	3. **Mechanism-of-Action Elucidation for Multi-Component Products**
	  ○ Molecular-level investigation of pharmacological mechanisms using advanced analytical and omics platforms
	  ○ Production of mechanistic data required to support regulatory acceptance of botanical drug products

Key Words	1. *Cordyceps militaris* 2. Botanical drug 3. Cordycepin 4. FDA IND clearance 5. Respiratory diseases

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	**Final Goal** Completion of global preclinical development for a *Cordyceps militaris*–derived botanical drug candidate (and/or a cordycepin-based natural product–derived synthetic drug) and submission of an FDA IND package to achieve IND effectiveness/clearance to initiate clinical trials. • Standardization of the manufacturing process and CMC package in compliance with the FDA's *Botanical Drug Development Guidance* and/or applicable guidance for synthetic drugs • Elucidation of multi-pronged pharmacological mechanisms of action (MOA) for *Cordyceps militaris* extract and cordycepin • Efficacy evaluation using clinically relevant animal models • Conduct of FDA-compliant GLP toxicology studies and acquisition of supporting results • Securing global Intellectual Property rights for the standardized cultivation process and drug candidate to establish global technological competitiveness • Execution of an FDA Pre-IND meeting and submission/filing of the final IND application			

	package
R&D Target Specifications	

Key Technology / Product Performance Indicator		Target	Remarks
1	Marker compound content	Cordycepin ≥ 9,000 ppm	Use a validated analytical method
2	Raw material standardization	Inter-batch variability across three batches within ±5%	Meet FDA CMC requirements
3	Efficacy assessment (respiratory diseases)	One efficacy report suitable for FDA IND submission	Study report based on animal efficacy models (e.g., COPD/asthma/lung cancer)
4	Safety assessment (toxicology)	One GLP nonclinical toxicology report suitable for FDA IND submission	Conducted at an FDA-recognized GLP facility (NOAEL to be established)
5	Heavy metal control	One residual heavy metal test report (CoA)	Residual heavy metal analysis in accordance with FDA requirements
6	Regulatory milestone	One FDA IND submission and filing/acceptance	-

R&D Plan

Manufacturing and CMC Development

- Optimization of extraction and purification processes to ensure high cordycepin content
- Establishment of validated analytical methods (HPLC/LC–MS)
- Pilot-scale formulation development for nonclinical and clinical studies

Smart-Farm–Based Raw Material Production

- Large-scale cultivation under controlled environmental conditions
- Implementation of GACP-compliant traceability from culture to harvest
- Minimization of inter-batch variability through environmental control

Efficacy and Mechanism-of-Action Studies

- In vitro pharmacology using cell-based assays
- In vivo efficacy studies in disease-relevant animal models
- Molecular-level MOA studies addressing inflammation, airway hyperresponsiveness, and cancer cell death

Regulatory Readiness and IND Submission

- Completion of GLP toxicology and safety pharmacology studies
- Preparation of CMC and nonclinical IND modules
- Conduct of Pre-IND meeting and IND submission

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Conduct globally standardized efficacy and safety evaluations • Perform molecular-level MOA studies for multi-component natural products • Execute GLP-compliant toxicology studies
Expected R&R of Korean SME Partner	• Large-scale cultivation and supply of standardized *Cordyceps militaris* materials • Establishment of marker-compound analytics and stability studies • Formulation development for clinical application • Overall project management, budgeting, and IND documentation coordination

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Impact** • Development of an innovative botanical drug candidate for respiratory diseases • Establishment of pharmaceutical-grade standardization technologies for natural products • Advancement of agricultural bio-resources into high-value therapeutic materials **Economic and Industrial Impact** • Entry into global botanical drug and respiratory therapeutics markets • Creation of licensing and co-development opportunities following IND clearance • Increased economic value of national agricultural resources **Societal Impact** • Strengthening of national capabilities in biopharmaceutical regulatory science • Contribution to treatment options for patients with unmet needs in respiratory diseases • Support for sustainable growth of the green bio industry

6. References

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